

CrabTransformer Training Report

Scenario: REAL | Loss: Tweedie | Lag: 0 Years | Current Spawner: Yes

— Training Configuration —

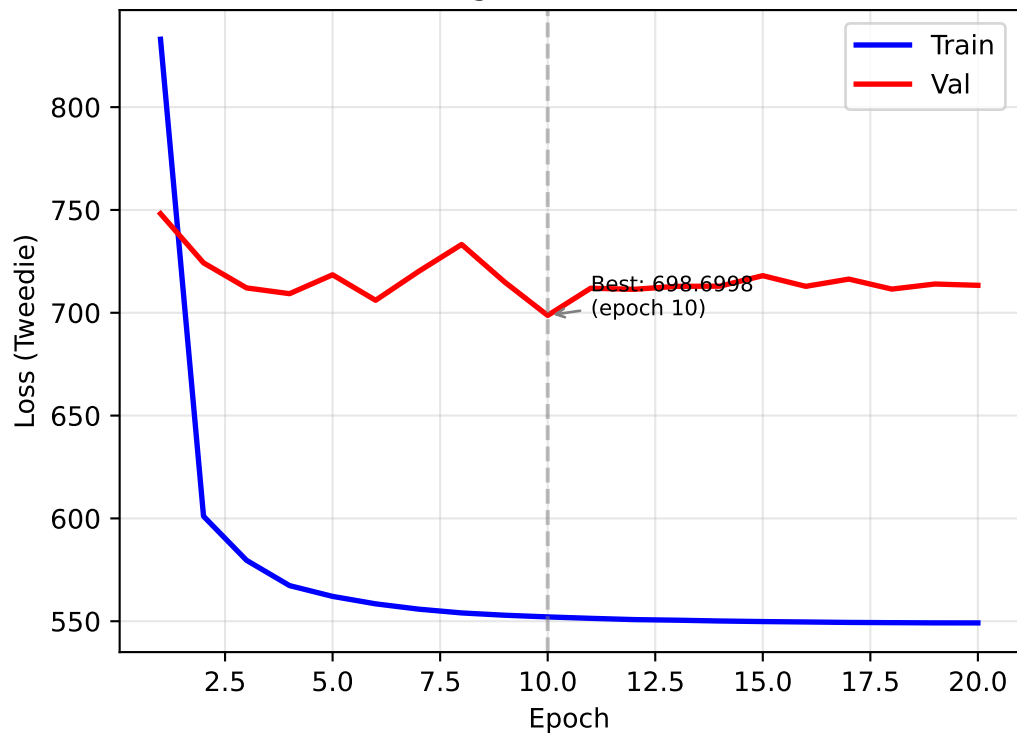
epochs_run: 20 best_val_epoch: 10
best_val_loss: 698.6998 final_train_loss: 549.1777
bias_correction (MSE): 1.0000
include_current_spawner: True

— Model Architecture —

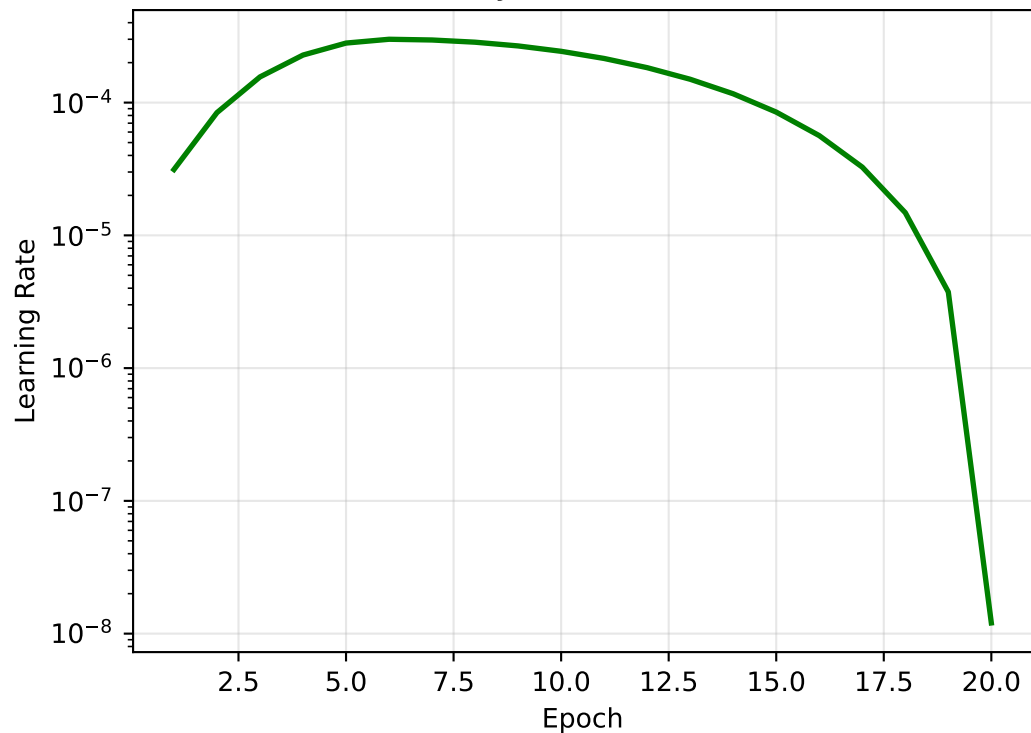
embed_dim: 128 num_heads: 4 num_layers: 3
d_ff: 512 dropout: 0.2 patch_size: 5
train/val/test years: 24/8/4

Training Curves: REAL | Lag 0 | Tweedie | w/ Spawner(t)

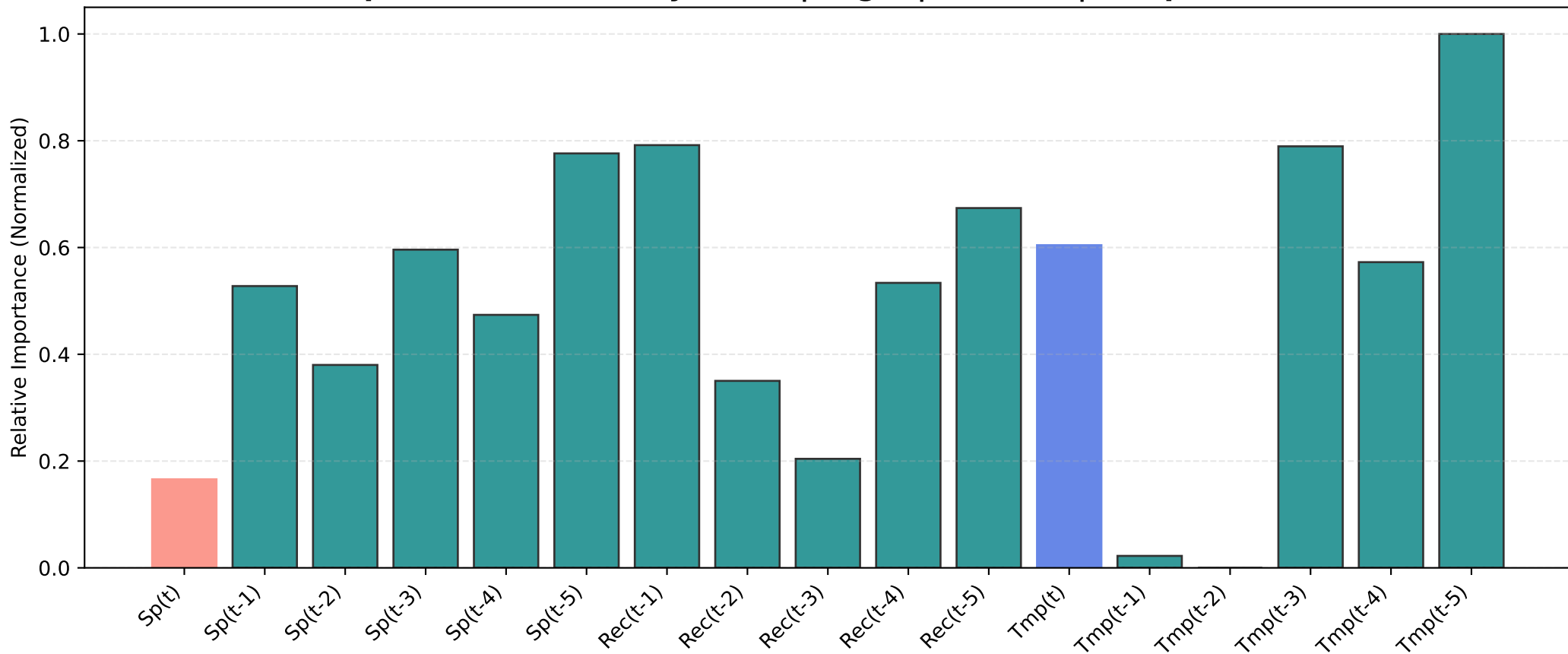
Training & Validation Loss



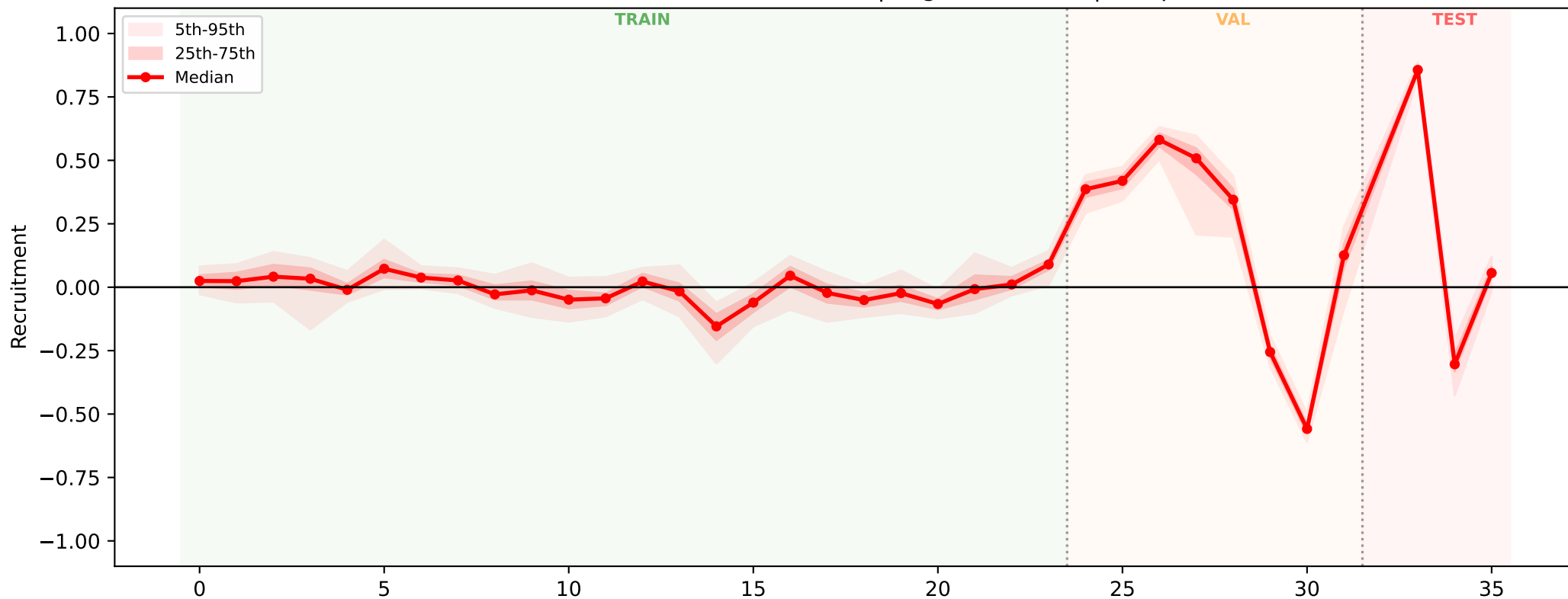
OneCycleLR Schedule



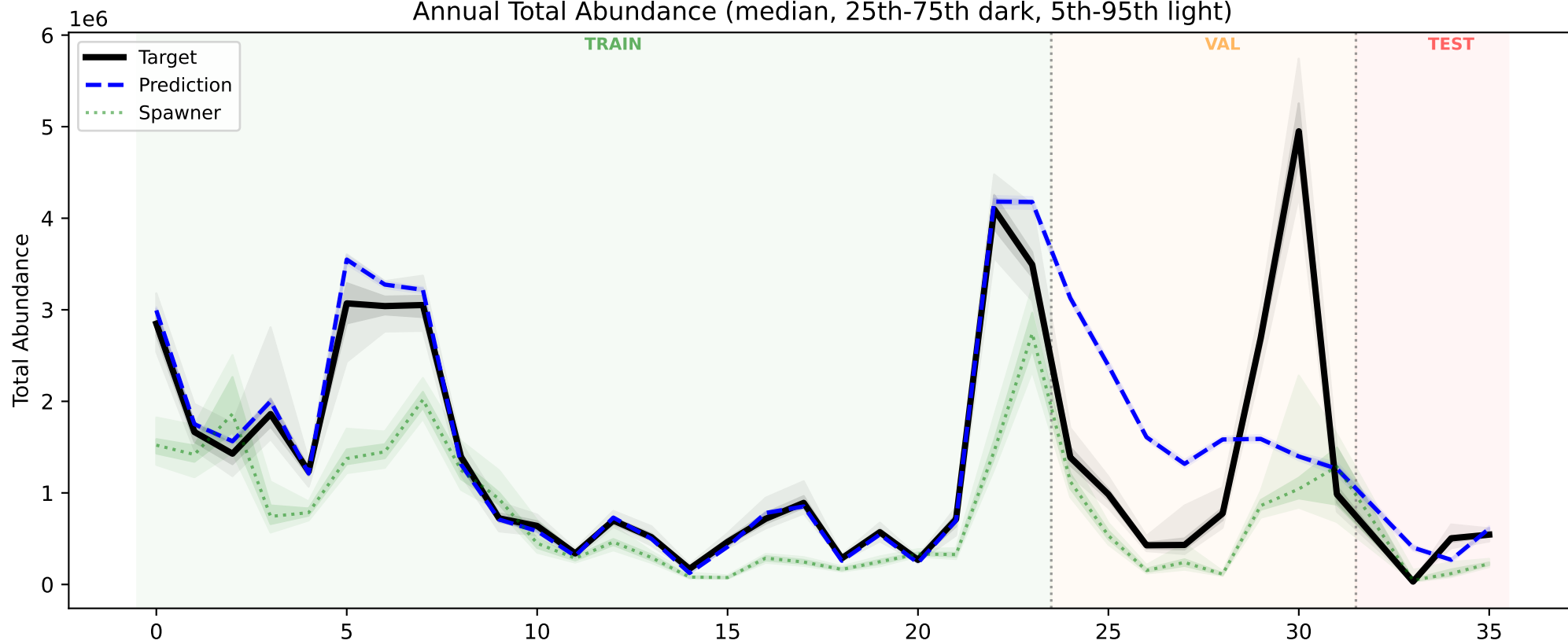
Input Channel Priority: REAL | Lag 0 | Tweedie | w/ Spawner(t)



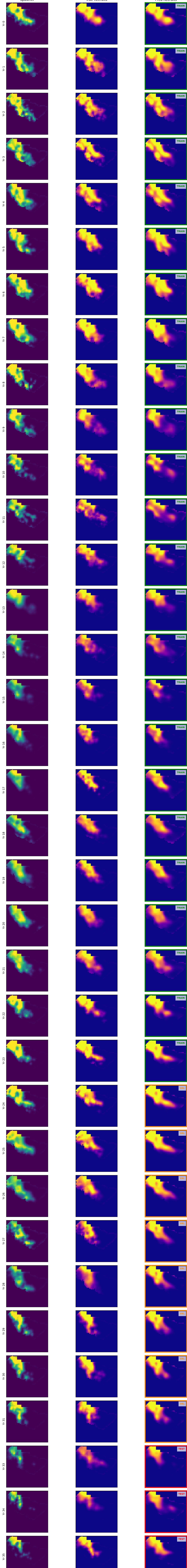
Annual Recruitment Deviations: REAL | Lag 0 (Tweedie) | w/ Spawner(t)



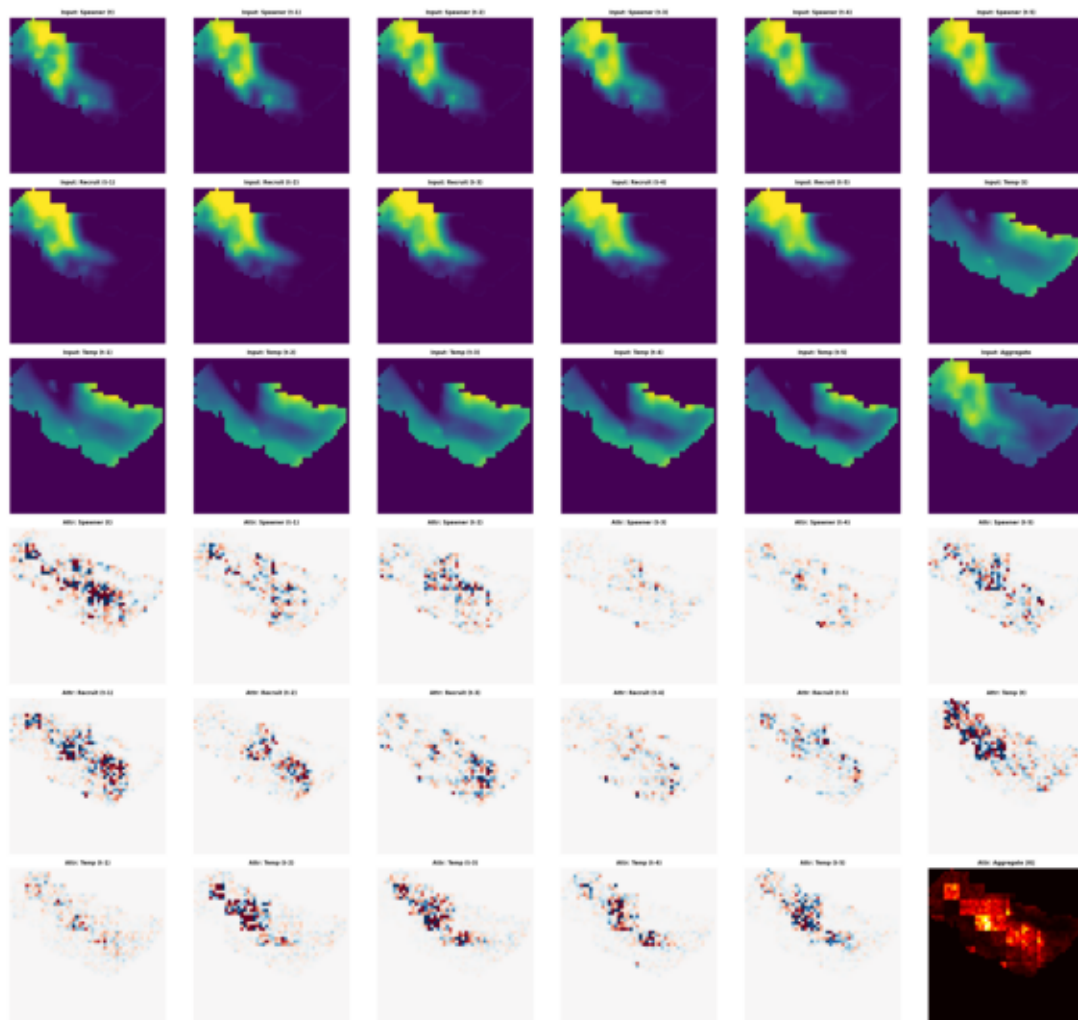
Annual Total Abundance (median, 25th-75th dark, 5th-95th light)



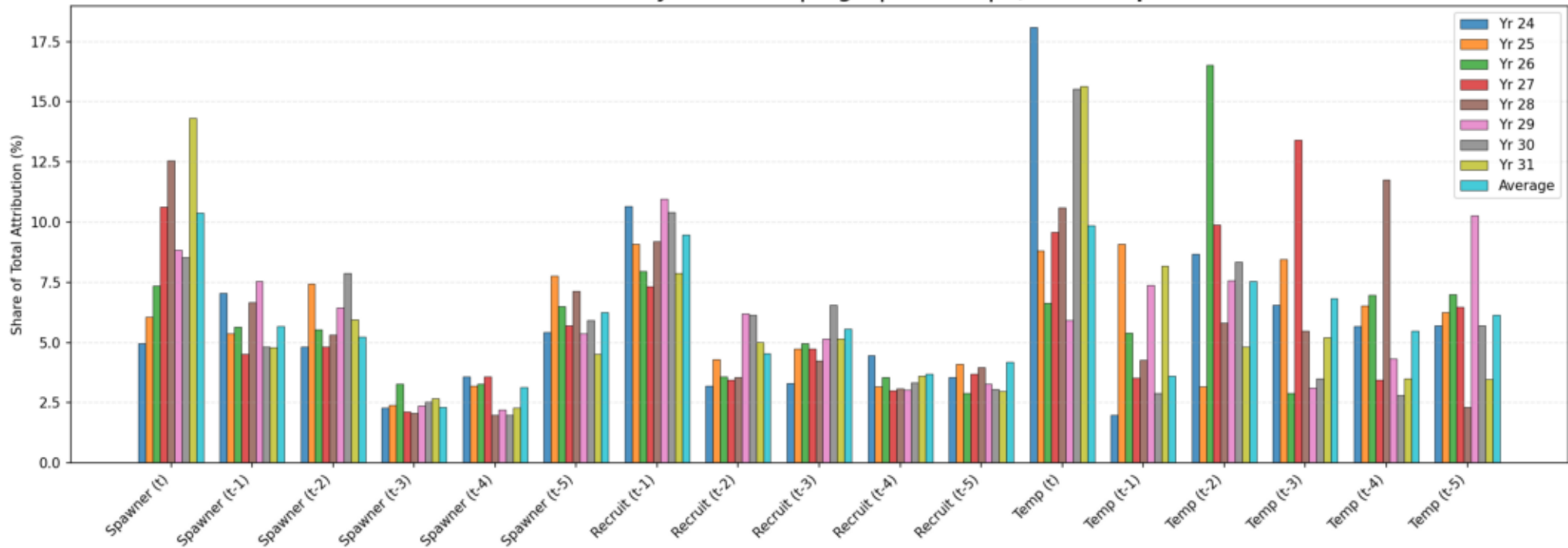
Spatial Trajectory: REAL | Lag 0 | Tweedie | w/ Spawner(t)



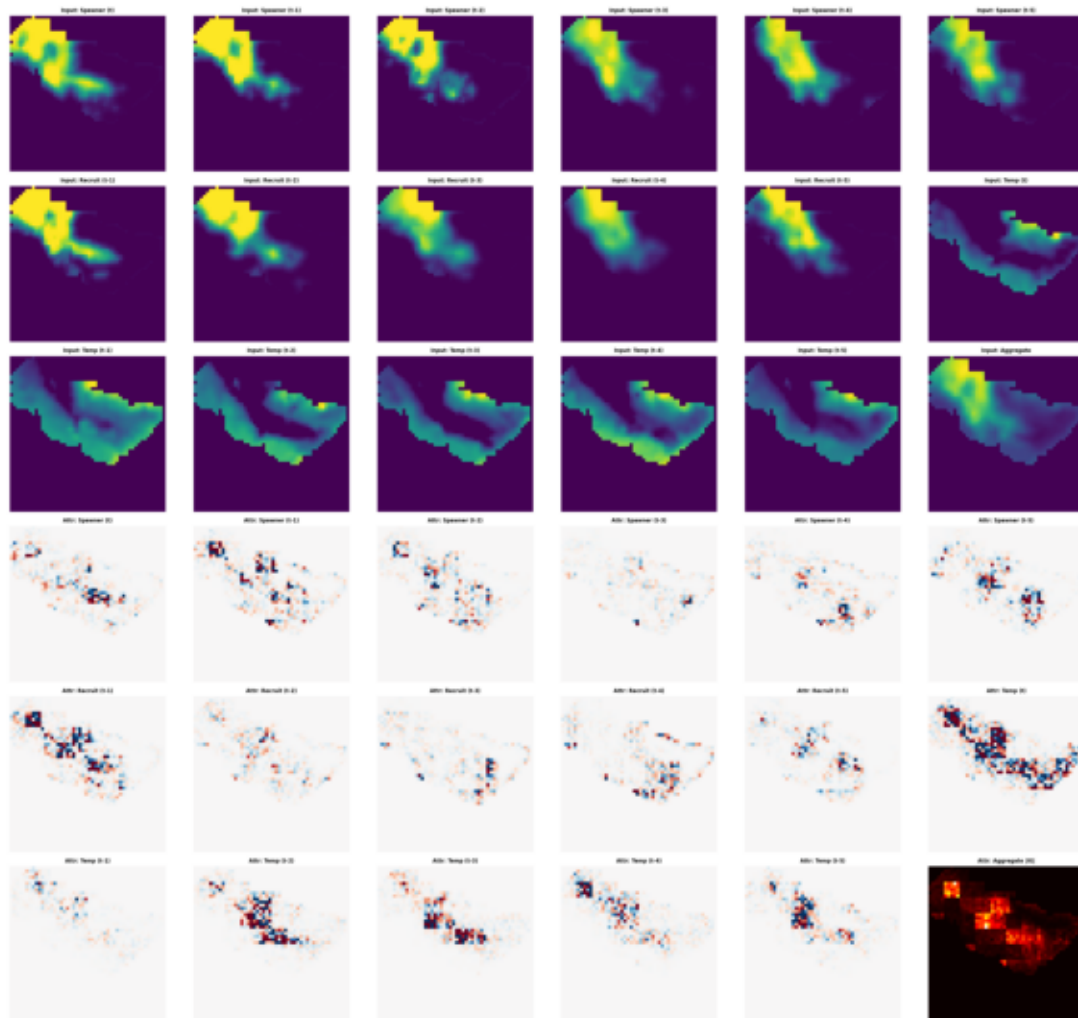
Year AVG — REAL | Lag 0 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



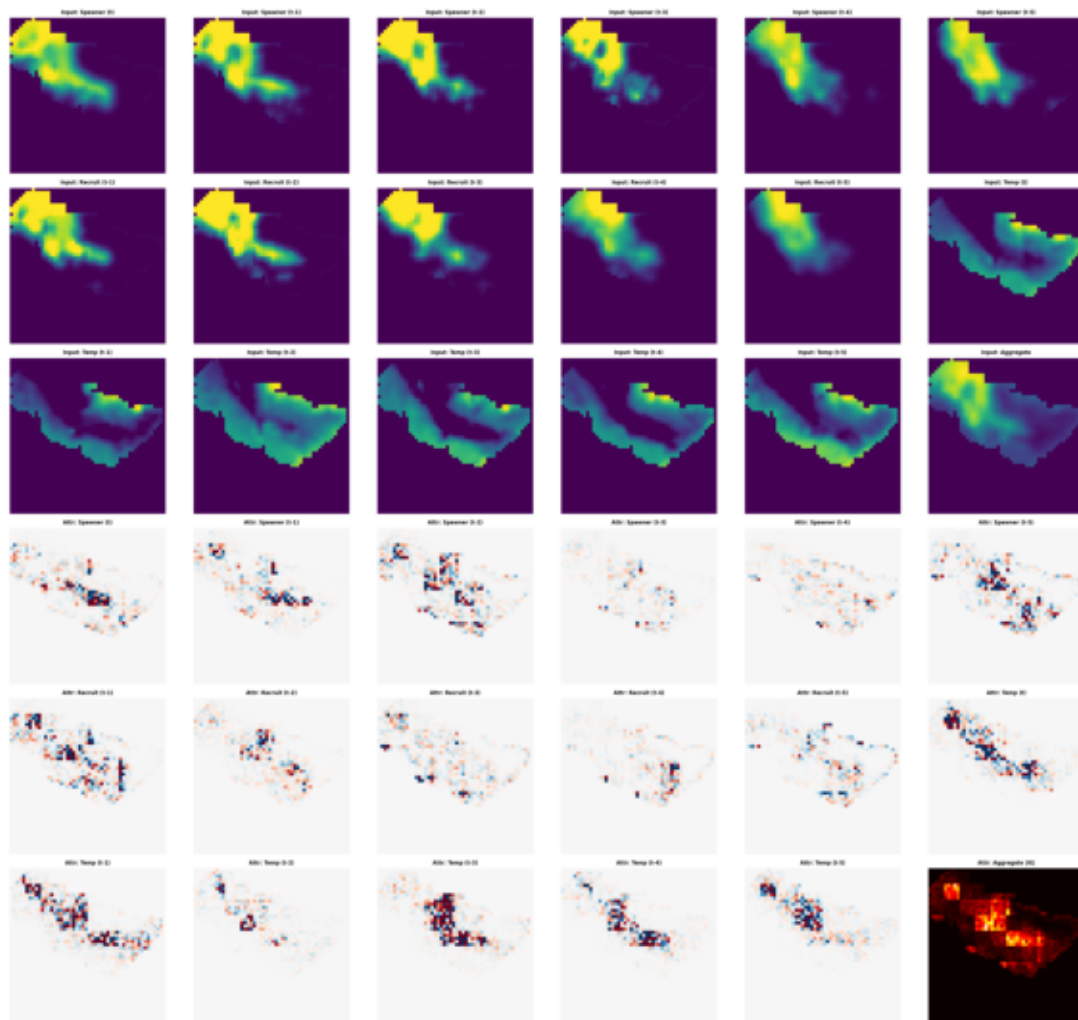
Channel Attribution by Year: REAL | Lag 0 | Tweedie | w/ Current Spawner



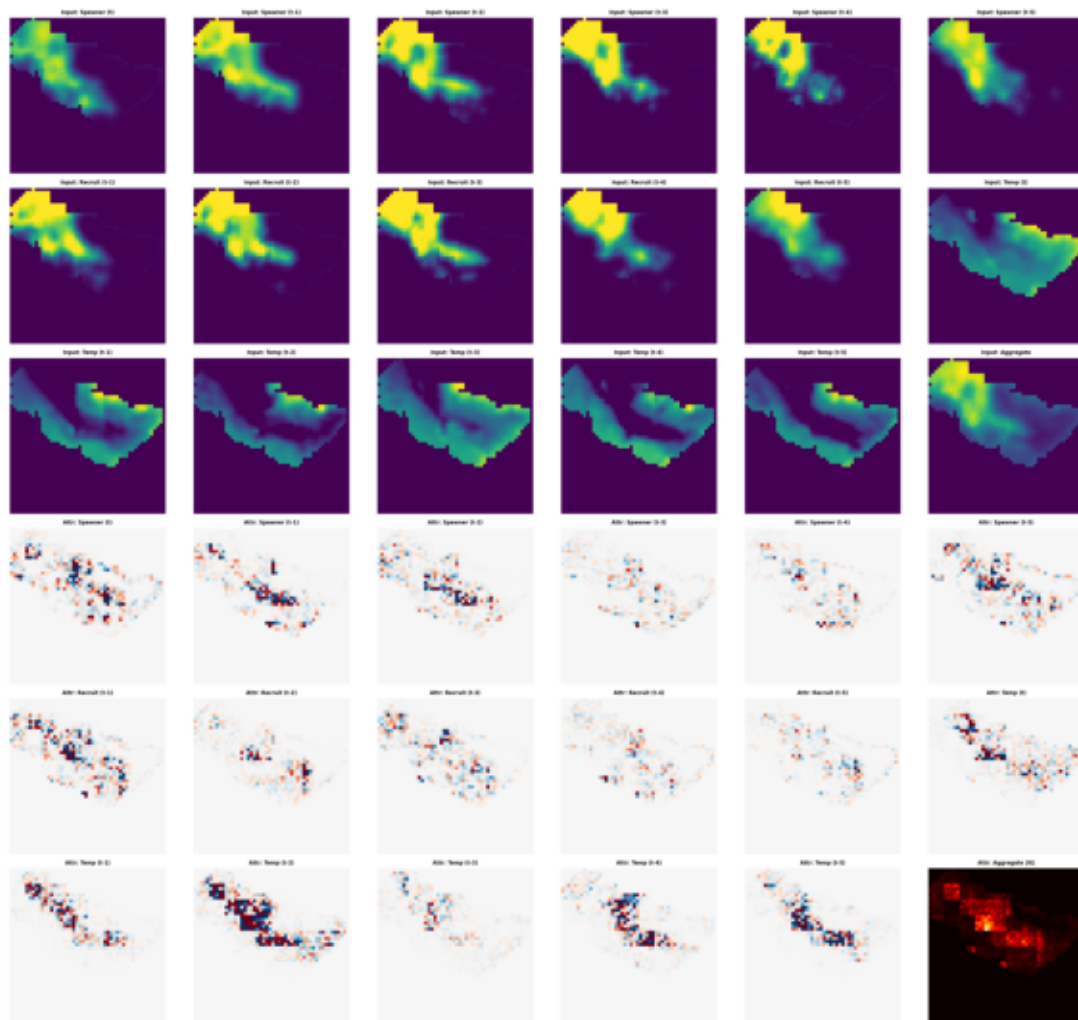
Year 24 — REAL | Lag 0 | Tweedie | w/ Current Spawner
Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



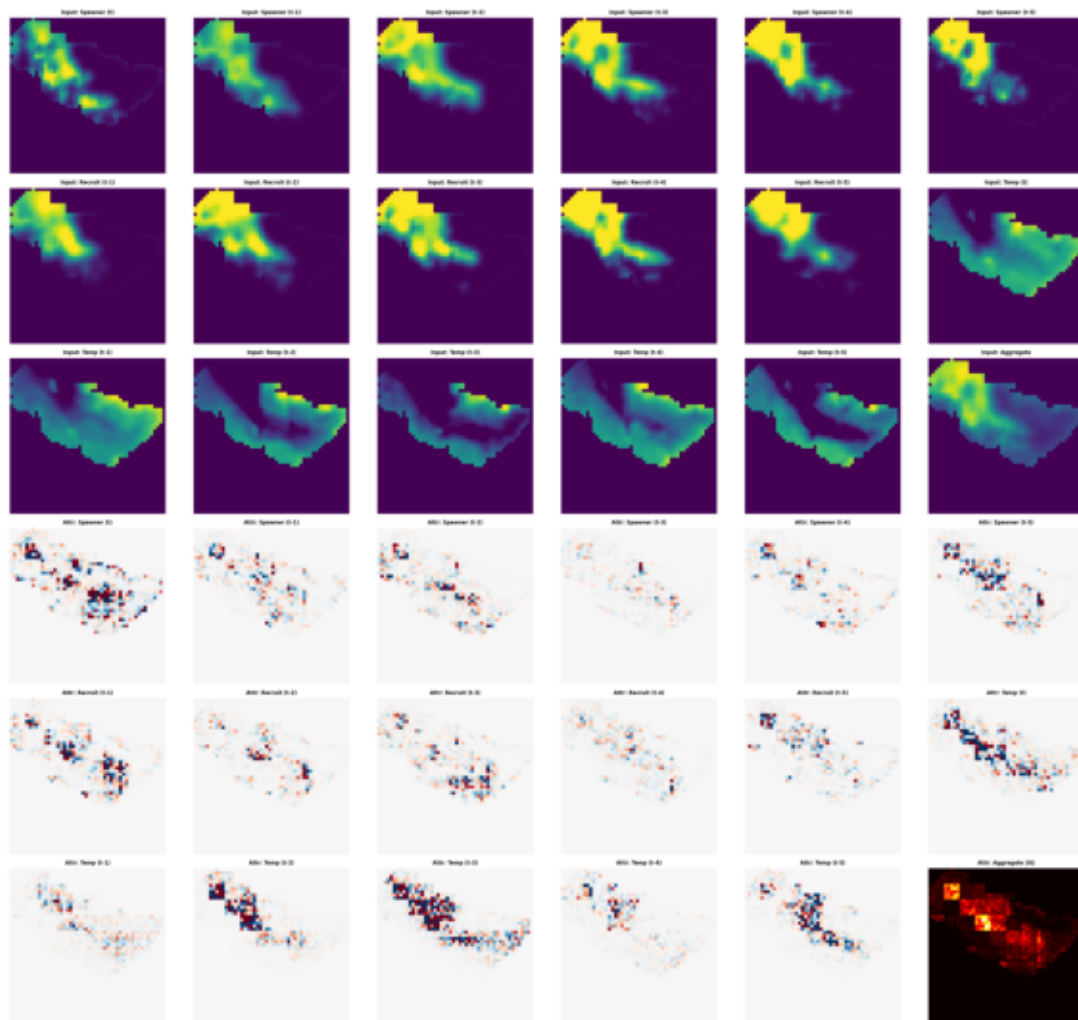
Year 25 -- REAL | Lag 0 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



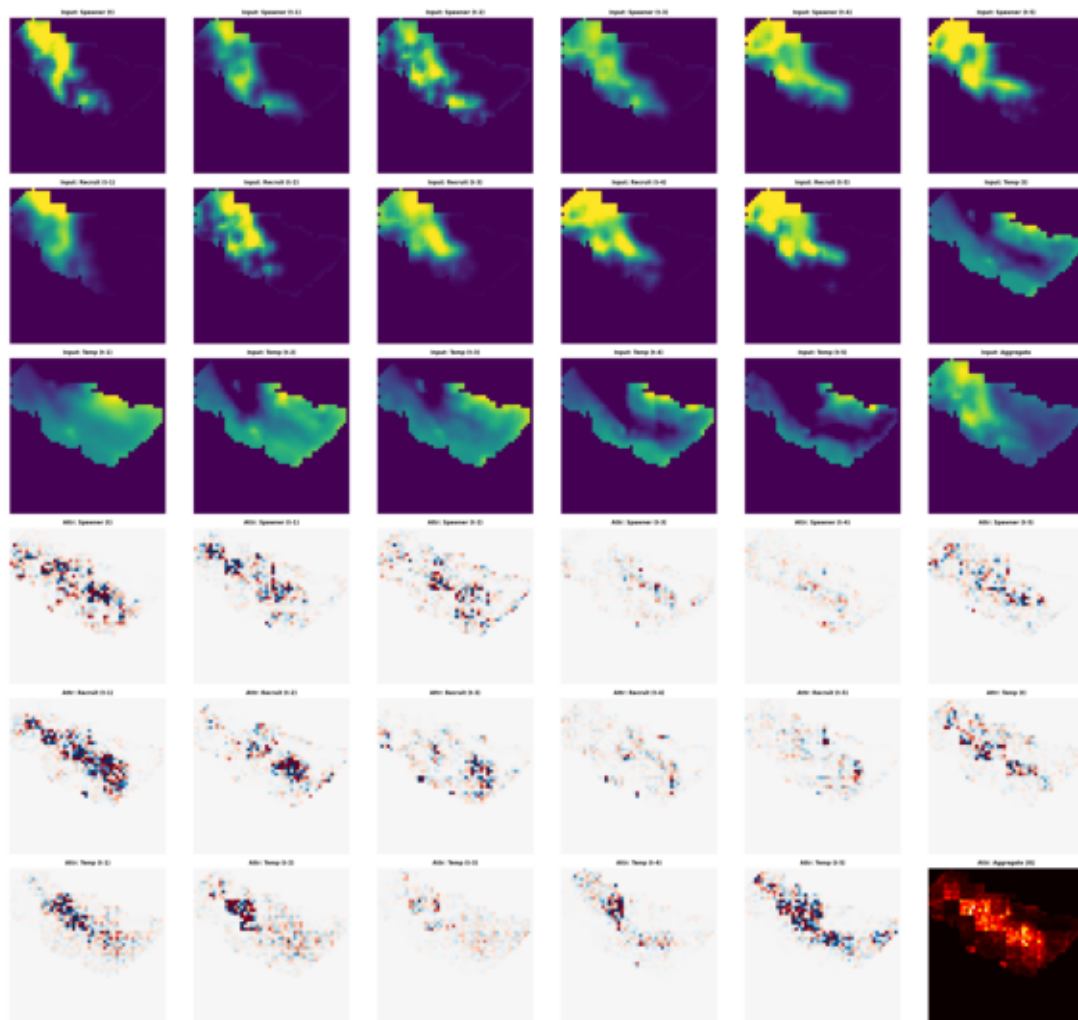
Year 26 -- REAL | Lag 0 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



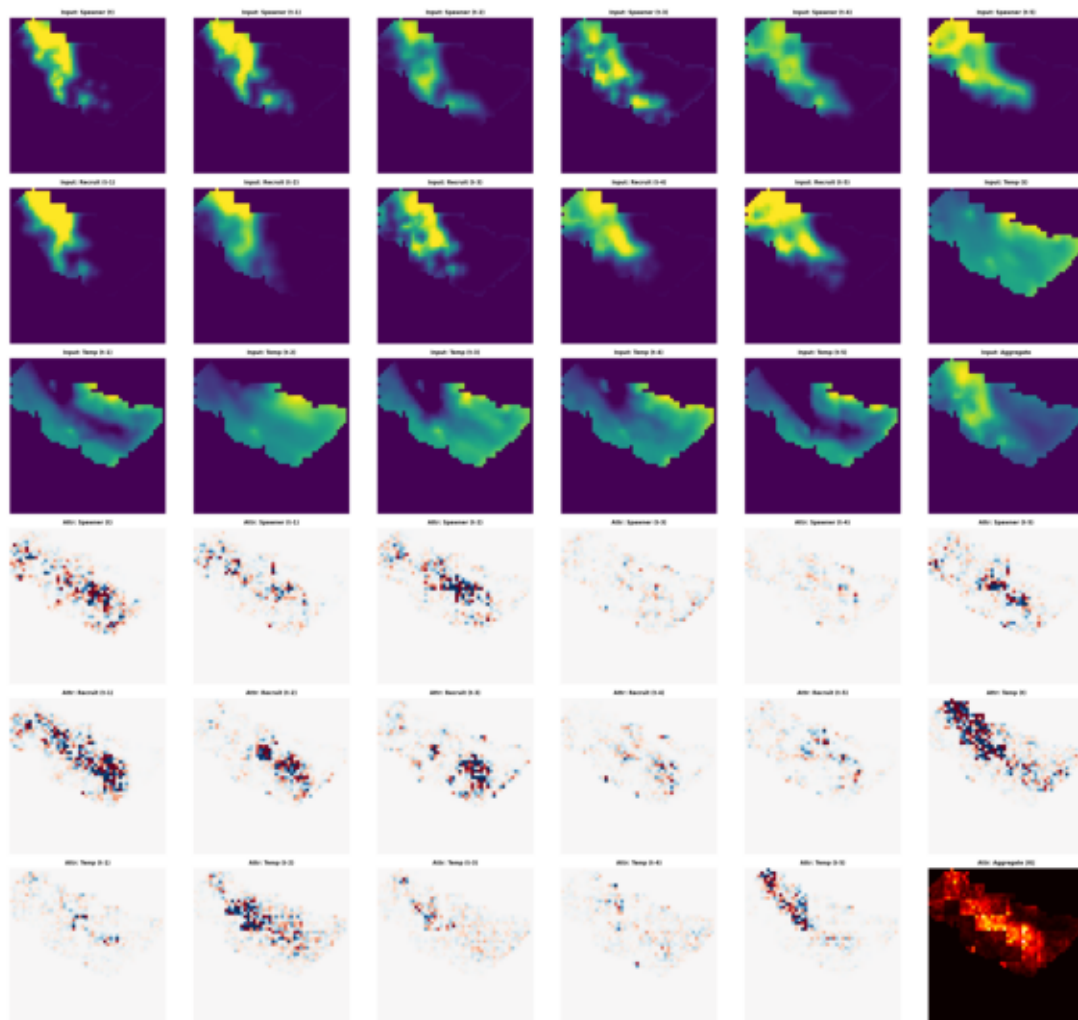
Year 27 -- REAL | Lag 0 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



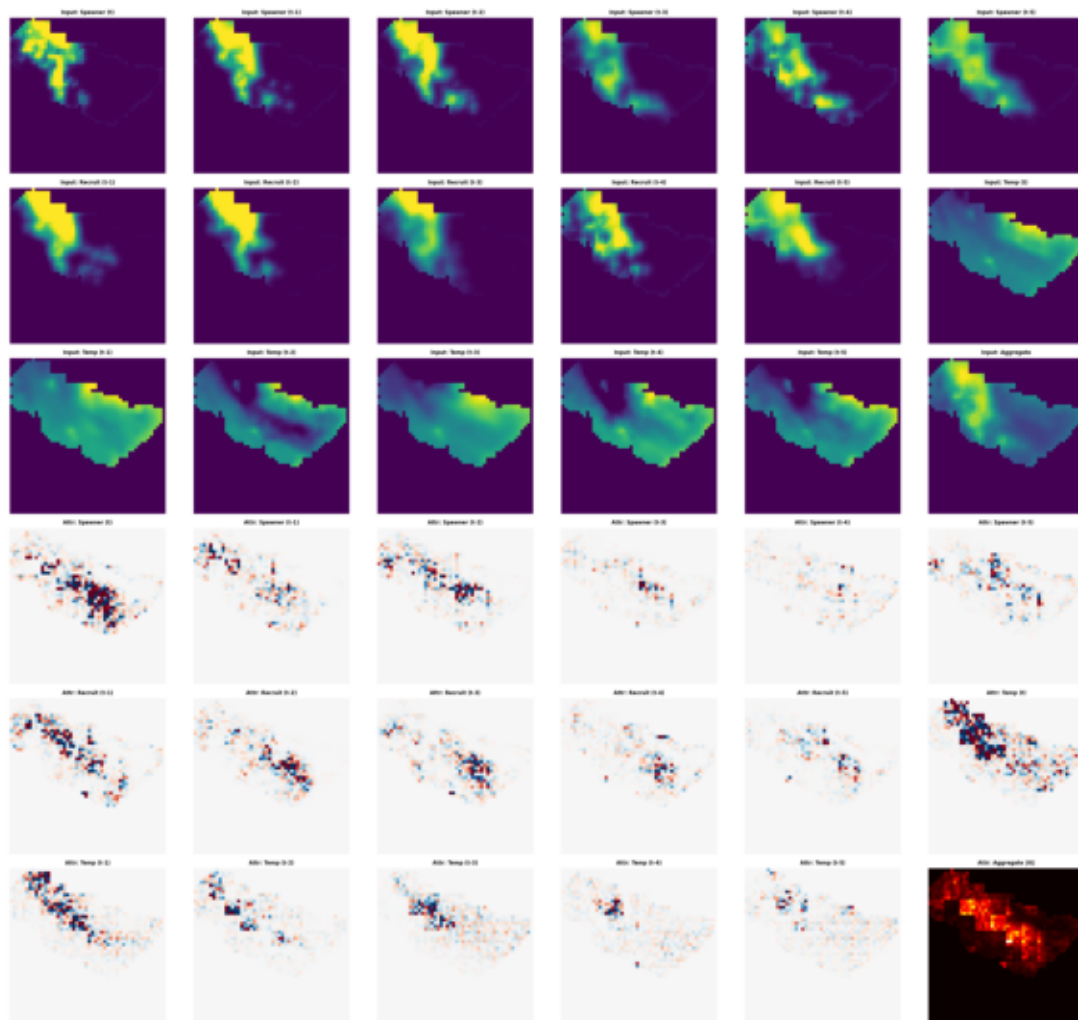
Year 29 -- REAL | Lag 0 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 30 -- REAL | Lag 0 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 31 — REAL | Lag 0 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Summary Statistics: REAL | Lag 0 | Tweedie | w/ Spawner(t)

Phase	Pred_Mean	Tgt_Mean	Pred_SD	Pred_P50	Pred_P95	Abundance_Capture	MAE	Spearman
TRAIN	1690.0	1612.8	6315.4	0.0	9824.9	1.048	408.0	0.955
VAL	2014.0	1801.5	5405.3	0.0	15146.7	1.118	2141.6	0.854
TEST	479.8	406.3	1558.3	0.0	3436.6	1.181	368.3	0.876

Detailed Metrics (R-style): REAL | Lag 0 | Tweedie | w/ Spawner(t)

Phase	RMSE	RMSE_unbias	Bias	Bias_%	Spearman	Pearson	Pct_Error	Zero_F1	TopK_Acc	TopK_Jaccard	Quant_Err	Skill_MSE	Skill_MAE
TRAIN	854.9	843.5	77.1	2.1%	0.963	0.983	-1.1%	0.962	0.924	0.862	0.26	0.955	0.878
VAL	6577.1	6380.8	212.5	6.3%	0.857	0.596	90.2%	0.912	0.580	0.416	0.89	-2.536	-0.055
TEST	1543.9	1518.6	73.4	3.2%	0.887	0.724	399.9%	0.957	0.705	0.562	0.55	-94.832	-1.964